- ***Reactive IDS:*** This detects malicious activity, alerts the administrator of the threats and also responds to those threats.

Numerous open source tools are available for enterprise networks, depending on the level of sophistication and security desired. In order to make the network highly secure, an IDS/IPS system should detect all sorts of suspicious activities coming to/from hosts in the network, and should take combative measures to prevent the attack.

Top 8 open source network intrusion detection tools

Here is a list of the top 8 open source network intrusion detection tools with a brief description of each.

Snort

Snort is a free and open source network intrusion detection and prevention tool. It was created by Martin Roesch in 1998. The main advantage of using Snort is its capability to perform real-time traffic analysis and packet logging on networks. With the functionality of protocol analysis, content searching and various pre-processors, Snort is widely accepted as a tool for detecting varied worms, exploits, port scanning and other malicious threats. It can be configured in three main modes -- sniffer, packet logger and network intrusion detection. In sniffer mode, the program will just read packets and display the information on the console. In packet logger mode, the packets will be logged on the disk. In intrusion detection mode, the program will monitor real-time traffic and compare it with the rules defined by the user.

Snort can detect varied attacks like a buffer overflow, stealth port scans, CGI attacks, SMB probes, OS fingerprinting attempts, etc. It is supported on a number of hardware platforms and operating systems like Linux, OpenBSD, FreeBSD, Solaris, HP-UX, MacOS, Windows, etc.

Pros:

- Free to download and is open source.
- Easy to write rules for intrusion detection.
- Highly flexible and dynamic in terms of live deployments.
- Good community support for solving problems and is under rapid development.

Cons:

- No GUI interface for rule manipulation.
- Somewhat slow in processing network packets.
- Cannot detect a signature split over multiple TCP packets, which occurs when packets are configured in *inline* mode.

Latest version: 2.9.9.0
Official website: *www.snort.org*

Security Onion

Security Onion is a Linux distribution for intrusion detection, network security monitoring and log management. The open source distribution is based on Ubuntu and comprises lots of IDS tools like Snort, Suricata, Bro, Sguil, Squert, Snorby, ELSA, Xplico, NetworkMiner, and many others. Security Onion provides high visibility and context to network traffic, alerts and suspicious activities. But it requires proper management by the systems administrator to review alerts, monitor network activity and to regularly update the IDS based detection rules.

Security Onion has three core functions:

- Full packet capture
- Network based and host based intrusion detection systems
- Powerful analysis tools

Full packet capture: This is done using netsnifff-ng, which captures all network traffic that Security Onion can see, and stores as much as your storage solution can hold. It is like a real-time camera for networks, and provides all the evidence of the threats and malicious activities happening over the network.

Network-based and host-based IDS: It analyses the network or host systems, and provides log and alert data for detected events and activity. Security Onion has varied IDS options like rule-driven IDS, analysis-driven IDS, HIDS, etc.

Analysis tools: In addition to network data capture, Security Onion comprises various tools like Sguil, Squert, ELSA, etc, for assisting administrators in analysis.

Security Onion also provides diverse ways for the live deployment of regular standalone, server-sensor and hybrid monitoring tools.

Pros:

- Provides a highly flexible environment for users to tune up network security as per the requirements.
- Consists of pre-installed sensor management tools, traffic analysers and packet sniffers, and can be operated without any additional IDS/IPS software.
- Has regular updates to improve security levels.

Cons:

- Doesn't work as an IPS after installation, but only as an IDS, and the user cannot find any instructions regarding this on the website.
- Doesn't support Wi-Fi for managing the network.
- Additional requirement for admins to learn various tools to make efficient use of the Security Onion distribution.
- No automatic backups of configuration files except rules; so usage of third party software is required for this activity.

Latest version: 14.04.5.1
Official website: *https://securityonion.net/*

OpenWIPS-NG

OpenWIPS-NG is a free wireless intrusion detection and prevention system that relies on sensors, servers and